Alamein International University
Computer Science & Engineering

EduVR: AI-Powered Collaborative Virtual Reality Learning Platform

A project submitted
in partial fulfillment of the requirements for the BSc. degree

by

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UNDERTAKING

This is to declare that the project entitled

"EduVR: AI-Powered Collaborative Virtual Reality Learning Platform "

is an original work done by the undersigned, in partial fulfillment of the requirements for the Bachelor of Science (BSc.) degree at the Faculty of Computer Science & Engineering (CSE), Alamein International University (AIU).

All the analysis, design, implementation, and system development activities have been accomplished by the undersigned. Moreover, this project has not been submitted to any other college or university for the award of any degree or diploma.

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ABSTRACT

Traditional educational methods often face significant challenges in effectively explaining complex three-dimensional concepts due to limited visualization, lack of interaction, and restricted access to physical laboratories. These limitations negatively impact students' understanding, engagement, and overall learning efficiency, especially in scientific and medical education.

This graduation project presents ***EduVR****, an AI-powered collaborative Virtual Reality learning platform designed to overcome these challenges by providing an immersive and interactive learning environment. The proposed system enables instructors and students to join shared virtual spaces where they can collaboratively explore high-quality 3D models in real time. EduVR supports advanced interactions such as rotation, isolation, and cross-sectional slicing of 3D objects, allowing learners to better understand internal structures and spatial relationships.*

In addition, the system integrates a ***Retrieval-Augmented Generation (RAG) based AI Smart Tutor*** *that allows students to ask questions using text or voice and receive accurate, context-aware responses derived from verified educational resources. EduVR combines a Unity-based multi-user VR environment, a web-based 3D visualization interface, and an automated 3D asset processing pipeline to ensure scalability, compatibility, and optimal performance.*

The proposed platform provides a cost-effective and scalable solution that enhances collaboration, improves conceptual understanding, and supports modern educational needs across various domains such as medicine, engineering, and science.

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1 Requirements Listing and Analysis

Requirements Analysis

This section defines the system requirements of the EduVR platform based on the identified educational needs and problem analysis. The requirements are classified into functional and non-functional requirements to clearly describe the expected behavior and quality attributes of the system.

Functional Requirements of the EduVR System

ID	Requirement Description
FR1	The system shall allow instructors and students to join shared virtual learning sessions.
FR2	The system shall support real-time multi-user interaction within a virtual environment.
FR3	The system shall allow instructors to control and manipulate shared 3D models.
FR4	The system shall allow students to view and interact with 3D models.
FR5	The system shall provide an AI Smart Tutor to answer user questions.
FR6	The system shall support text and voice-based interaction with the AI tutor.

2 Solution Design and Diagrams

Solution Design

The EduVR solution is designed as a collaborative learning platform that integrates Virtual Reality, Artificial Intelligence, and real-time networking technologies. The system design focuses on modularity and scalability to allow future extensions and support different educational domains.

The system architecture is divided into user interfaces, backend services, AI services, and data storage components. The design is illustrated using architecture and workflow diagrams that demonstrate the interaction between users, system components, and AI services. These diagrams provide a clear representation of data flow, system responsibilities, and component communication.

3 Project Planning

Project Planning

The development of the EduVR project followed a structured software engineering approach to ensure timely delivery and system quality. The project was divided into several phases, including requirements analysis, system design, implementation, testing, and evaluation.

The initial phase focused on defining project objectives and selecting appropriate technologies. The design phase involved planning the system architecture and interaction flow. The implementation phase focused on developing the VR environment, backend services, and AI components. Finally, testing and evaluation were conducted to validate system functionality and performance.

Team roles were distributed among members to ensure efficient collaboration. Each team member was responsible for specific tasks such as development, documentation, research, and testing. This planning approach helped ensure clear responsibilities and smooth project progress.

4 Implementation Approach

Implementation Approach

The implementation of EduVR focused on integrating Virtual Reality interaction, multi-user collaboration, and AI-based assistance into a unified system. Unity was used to develop the VR environment and support user movement and object interaction. Real-time collaboration was implemented to synchronize user actions and shared objects.

Backend services were implemented to handle authentication, session management, and communication between system components. The AI Smart Tutor was implemented using a Retrieval-Augmented Generation approach, where user queries are processed by retrieving relevant information from verified sources before generating responses.

An automated pipeline was implemented to process and optimize 3D assets for use across both VR and web platforms. This approach ensured consistent performance and reduced manual preparation effort.

5 Adopted Ethics Principles

Adopted Ethics Principles

The EduVR project was developed in accordance with recognized software engineering ethics principles. The primary ethical principles adopted include user privacy, data security, transparency, and responsible use of Artificial Intelligence.

User data and interactions are handled securely, and role-based access control is implemented to prevent unauthorized actions. The AI Smart Tutor follows responsible AI practices by using verified knowledge sources to reduce incorrect or misleading information. Ethical alternatives such as using unrestricted AI models were considered; however, a Retrieval-Augmented Generation approach was chosen to ensure accuracy and reliability.

All software tools, libraries, and frameworks used in the project follow their respective licenses and usage policies. Open-source tools were used in compliance with their licenses, and no copyrighted content was used without permission. User privacy is respected by limiting data collection to essential information required for system operation.

References

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